Pairs Trading via Ornstein–Uhlenbeck Mean Reversion

A Continuous-Time Stochastic Calculus Approach

Meemansa Jha- 2024B4A71096G Vedika Jain- 2024B4A81111G

Stochastic Calculus and its Applications in Finance

April 27, 2026

Motivation: The Pairs Trading Idea

Two stocks in the same industry share most of their daily moves.

Observation:

- Each individual price is non-stationary – it drifts off to infinity.
- Yet the *spread* between two cointegrated stocks wobbles around a stable mean.

Strategy:

- When the spread drifts unusually far from its mean, bet on reversion.
- Short the relatively expensive leg, long the cheap leg, close when it returns.

Mathematical model: the spread follows an Ornstein–Uhlenbeck (OU) process – the simplest mean-reverting Itô SDE, structurally identical to Shreve's Vasicek interest-rate model.

Why Is the Spread Stationary?

Each log-price is a common factor + idiosyncratic noise:

$$\log S_A(t) = c_A + \beta\, F(t) + \xi_A(t)$$
$$\log S_B(t) = c_B + \; F(t) + \xi_B(t)$$

where $F(t)$ = common (macro/sectoral) factor – non-stationary; ξ_i = idiosyncratic noise – stationary.

Form the spread $Z(t) = \log S_A(t) - \alpha - \beta \log S_B(t)$ (with $\alpha = c_A - \beta c_B$):

$$Z(t) \;=\; \xi_A(t) \;-\; \beta\, \xi_B(t)$$

The non-stationary common factor $F(t)$ *cancels* – only stationary noise remains.

Engle–Granger (1987): if such a β exists, the pair is *cointegrated*. The spread can then be modelled by a stationary, mean-reverting SDE.

The Ornstein–Uhlenbeck SDE

Stochastic differential equation

$$\mathrm{d}X(t) \;=\; \theta\big(\mu - X(t)\big)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W(t), \qquad X(0) = x_0$$

- $\theta > 0$ – **mean reversion speed**: how fast X is pulled back to μ.
- μ – **long-run mean**: the level X reverts to.
- $\sigma > 0$ – **volatility**: size of the random kicks.
- $W(t)$ – standard Brownian motion.

Reading the SDE:

- Drift $\theta(\mu - X)$ *always* points toward μ – a restoring force, like a spring.
- Diffusion $\sigma\,\mathrm{d}W$ adds random noise that prevents X from collapsing onto μ.

Same equation as Vasicek model: identify $\theta_{\text{ours}} = \beta_{\text{Vasicek}}$, $\mu_{\text{ours}} = a_{\text{Vasicek}}/\beta_{\text{Vasicek}}$.

Closed-Form Solution: Strategy

Goal: find an explicit formula for $X(t)$.

Strategy: *integrating factor*.
The drift contains the term $-\theta X(t)\,\mathrm{d}t$, which couples X to itself. We multiply X by a deterministic factor chosen to *cancel* this coupling.

Step 1: Define the auxiliary process

$$Y(t) \;:=\; e^{\theta t}\,X(t).$$

Step 2: Compute $\mathrm{d}Y(t)$ using the Itô–Doeblin formula with $f(t,x) = e^{\theta t}\,x$:

$$\begin{aligned} f_t &= \theta\, e^{\theta t}\, x \\ f_x &= e^{\theta t} \\ f_{xx} &= 0 \quad \text{(linear in } x\text{)} \end{aligned}$$

The $f_{xx} = 0$ is the key: *no Itô correction term* appears.

Closed-Form Solution: The Cancellation

Step 3: apply Itô–Doeblin and substitute the OU SDE.

$$\mathrm{d}Y(t) \;=\; f_t\,\mathrm{d}t + f_x\,\mathrm{d}X(t) + \tfrac{1}{2} f_{xx}\,(\mathrm{d}X)^2.$$

With $f_{xx} = 0$:

$$\mathrm{d}Y(t) \;=\; \theta\, e^{\theta t}\, X(t)\,\mathrm{d}t + e^{\theta t}\,\mathrm{d}X(t).$$

Substituting $\mathrm{d}X = \theta(\mu - X)\,\mathrm{d}t + \sigma\,\mathrm{d}W$:

$$\mathrm{d}Y(t) \;=\; \theta\, e^{\theta t}\, X(t)\,\mathrm{d}t + e^{\theta t}\left[\theta(\mu - X(t))\,\mathrm{d}t + \sigma\,\mathrm{d}W(t)\right].$$

The two $X(t)\,\mathrm{d}t$ terms cancel exactly:

$$\mathrm{d}Y(t) \;=\; \theta\mu\, e^{\theta t}\,\mathrm{d}t \;+\; \sigma\, e^{\theta t}\,\mathrm{d}W(t).$$

The transformed SDE has *deterministic* coefficients (no Y on the right-hand side), so it can be integrated directly.

Closed-Form Solution: The Result

Step 4: integrate from 0 to t and divide by $e^{\theta t}$.

The first term integrates by ordinary calculus to $\mu(e^{\theta t} - 1)$; the second is an Itô integral (Shreve §4.3) and stays as such:

$$Y(t) - x_0 \;=\; \mu\big(e^{\theta t} - 1\big) + \sigma \int_0^t e^{\theta s}\, \mathrm{d}W(s).$$

Substituting $Y(t) = e^{\theta t} X(t)$ and dividing:

Closed-form OU solution

$$X(t) \;=\; \underbrace{x_0\, e^{-\theta t}}_{\text{decay of } x_0} \;+\; \underbrace{\mu(1 - e^{-\theta t})}_{\text{pull toward } \mu} \;+\; \underbrace{\sigma \int_0^t e^{-\theta(t-s)}\, \mathrm{d}W(s)}_{\text{accumulated noise}}$$

The exponential kernel $e^{-\theta(t-s)}$ in the Itô integral **discounts old shocks**: recent noise gets full weight, old noise decays. The process has a *fading memory*.

Stationary Distribution & Half-Life

Mean and variance. Take expectations using the zero-mean property of Itô integrals; compute variance via the Itô isometry:

$$\mathbb{E}[X(t)] = x_0\, e^{-\theta t} + \mu\big(1 - e^{-\theta t}\big)$$

$$\mathrm{Var}[X(t)] = \frac{\sigma^2}{2\theta}\big(1 - e^{-2\theta t}\big)$$

As $t \to \infty$:

Stationary distribution

$$X(t) \implies \mathcal{N}\left(\mu,\ \frac{\sigma^2}{2\theta}\right) \quad \text{with stationary std dev} \quad \sigma_s = \frac{\sigma}{\sqrt{2\theta}}.$$

Half-life. Setting $\mathbb{E}[X(t)] - \mu = \frac{1}{2}(x_0 - \mu)$ gives

$$\tau = \frac{\ln 2}{\theta}.$$

This anchors the strategy: the z-score $Z(t) = (X - \mu)/\sigma_s$ is standard-normal in the stationary state. So $|Z| > 2$ is a $\sim 5\%$ tail event – rare and mean-reversion-worthy.

Methodology: Three Steps

Step 1 – Cointegration screen (Engle–Granger 2-step):

- OLS regress $\log S_A$ on $\log S_B \Rightarrow$ hedge ratio β, residuals.
- Augmented Dickey–Fuller test on residuals.
- Accept pair if ADF < -3.34 (5% MacKinnon critical value).

Step 2 – OU calibration (next slide).

Step 3 – Trading rule based on z-score (slide after).

Data and split.

- Yahoo Finance daily adjusted-close prices, 2021-01 to 2024-12.
- 8 candidate US/Indian pairs across 7 sectors.
- First 75% of trading days for fitting (*training*); last 25% for backtest (*out-of-sample*).
- No test data informs the screen, the fit, or the thresholds.

Calibration: From OU to AR(1)

The OU closed-form, applied over one timestep Δt, gives

$$X_{t+\Delta t} \;=\; X_t\, e^{-\theta\Delta t} + \mu\big(1 - e^{-\theta\Delta t}\big) + \sigma \int_t^{t+\Delta t} e^{-\theta(t+\Delta t - s)}\, \mathrm{d}W(s).$$

Defining $a := e^{-\theta\Delta t}$, $b := \mu(1-a)$, this is a **discrete-time AR(1)**:

$$X_{t+\Delta t} \;=\; a\,X_t \;+\; b \;+\; \eta_t, \qquad \eta_t \sim \mathcal{N}\big(0,\, \sigma^2(1-a^2)/(2\theta)\big).$$

Fitting. OLS regression of $X_{t+\Delta t}$ on X_t gives $\hat{a}, \hat{b}$. Invert:

$$\hat{\theta} = -\frac{\ln \hat{a}}{\Delta t}, \qquad \hat{\mu} = \frac{\hat{b}}{1-\hat{a}}, \qquad \hat{\sigma} = \sqrt{\frac{2\hat{\theta}\,\hat{V}_\eta}{1-\hat{a}^2}}.$$

With $\Delta t = 1/252$, parameters come out in per-year units.

Note: this discretisation is *exact*, not an approximation, because the OU SDE has a closed-form solution.

Trading Rule

Define the standardised spread

$$Z(t) \;=\; \frac{X(t) - \hat{\mu}}{\hat{\sigma}_s}.$$

In the stationary state $Z \sim \mathcal{N}(0,1)$.

Rule:

- **Enter** when $|Z| > 2$ (short spread if $Z > 2$, long if $Z < -2$).
- **Take profit** when $|Z| < 0.5$.
- **Stop-loss** when $|Z| > 3.5$.
- **Time-stop** after 2τ (two half-lives).
- **Re-arm** only when $|Z|$ re-enters the ± 2 band.

Justification of thresholds.

- ± 2: $\sim 5\%$ tail of stationary $\mathcal{N}(0,1)$ – statistically rare.
- ± 0.5: take most of the reversion without waiting for exact zero crossing.
- ± 3.5: $\sim 0.05\%$ tail – if we hit this, the model has likely failed.
- 2τ: by then $\mathbb{E}[|Z|]$ has decayed to 0.5 by theory.

Costs: 10 bp per leg per trade.

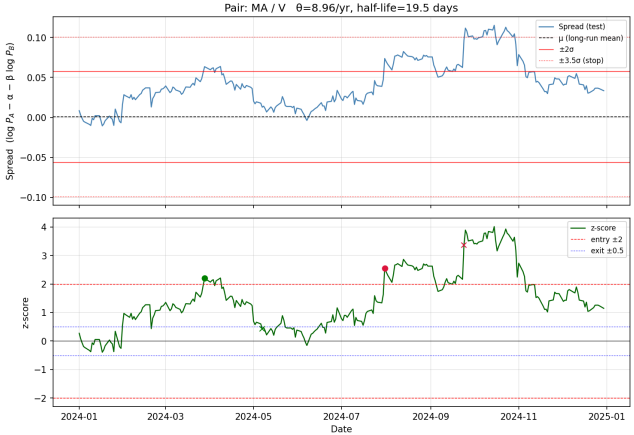
Results – Pair Screening

Pair	β	ADF	Coint?	θ/yr	σ_s	Half-life (d)
KO/PEP	0.738	−2.46	No	3.72	0.036	47.0
XOM/CVX	1.243	−1.76	No	1.92	0.082	91.1
MA/V	**0.952**	**−3.77**	**Yes**	**8.96**	**0.029**	**19.5**
GLD/SLV	0.231	−1.30	No	1.57	0.056	111.6
HD/LOW	0.839	−2.31	No	3.85	0.046	45.3
HDFCBANK/ICICIBANK	0.248	−3.15	No	6.31	0.055	27.7
TCS/INFY	0.467	−2.02	No	3.48	0.057	50.1
RELIANCE/ONGC	**0.336**	**−3.39**	**Yes**	**6.79**	**0.061**	**25.7**

Two of eight pairs pass the 5% test: **MA/V** (ADF $= -3.77$) and **RELIANCE/ONGC** (ADF $= -3.39$).

The other six fail despite belonging to the same sector – *sectoral similarity does not guarantee cointegration*. The screen is doing real work; 75% of obvious candidates are rejected.

Results – Out-of-Sample Backtest



MA/V spread (top) and z-score (bottom). Dots = entries, crosses = exits.

Backtest summary (Nov 2023 – Dec 2024)

# trades	3
Wins	2
Success ratio	**66.7%**
Total net P&L	**+2.07%**
Sharpe	0.53
Max drawdown	−2.67%

Trade 1 (Mar–May 24): enter at $z \approx +2.2$, monotone reversion, exit at $+0.5\sigma_s$. **+4.7%**.

Trade 2 (Aug–Sep 24): enter at $z \approx +2.5$, spread drifts to $+4\sigma_s$ – cointegration breakdown – stop-loss fires. −2.7%.

Discussion – What the Two Trades Teach Us

Trade 1: textbook OU reversion.

- Spread drifts to $+2.2\sigma_s$, reverts in $\sim$30 days (close to the estimated 19.5-day half-life).
- Behaviour matches theory's prediction almost exactly.
- When the stationarity assumption holds, the statistical edge is real.

Trade 2: temporary cointegration breakdown.

- Spread drifts to $+4\sigma_s$ instead of reverting – a $\sim 0.003\%$ event under the OU model. The model has failed in this window.
- The $3.5\sigma_s$ stop-loss limits the damage to -2.7%.
- Without the stop, the position would have stayed open in drawdown for nearly three more months.

Both behaviours are predicted: OU theory says deviations *should* revert in expectation, but does not say every realisation will. The strategy is a positive-expected-value bet, not a deterministic arbitrage.

Conclusion & Limitations

What worked.

- Cointegration filter has real teeth – 6 of 8 sector-matched candidates rejected.
- Both retained pairs profitable out-of-sample.
- Trade behaviour matches OU theory: clean reversion when stationarity holds, controlled stop-loss when it fails.

Honest limitations.

- **Sample size:** 3 trades only; 95% CI on hit rate is wide $[21\%, 94\%]$.
- **Survivorship bias:** candidates were hand-picked from today's well-known sector analogues.
- **Static β:** a Kalman-filtered β would adapt to drift and likely reduce stop-outs.
- **Cost sensitivity:** doubling to 20 bp/leg halves MA/V profit; turns RELIANCE negative.

Bottom line. The OU model – derived directly from the Itô–Doeblin formula – gives a coherent bridge from the continuous-time machinery of to a concrete, testable trading strategy. Modest but positive evidence, behaving exactly as theory predicts.

Thank you

References: Shreve (2004) Stochastic Calculus for Finance II; Engle & Granger (1987) Econometrica; Gatev, Goetzmann & Rouwenhorst (2006) RFS; MacKinnon (2010).